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Palaeomagnetism and $^{40}Ar/^{39}Ar$ age from a Cretaceous volcanic sequence, Inner Mongolia, China: Implications for the field variation during the Cretaceous normal superchron

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ABSTRACT

An integrated palaeomagnetic and $^{40}Ar/^{39}Ar$ dating study was carried out on an Early Cretaceous volcanic lava sequence from the Suhongtu section, Inner Mongolia, to determine the field behavior within the Cretaceous normal superchron (CNS). $^{40}Ar/^{39}Ar$ ages were obtained from 12 lava flows, indicating that the studied lava was formed around 114.1 ± 0.3 Ma for the lower interval and 110.6 ± 0.1 Ma for the upper interval. Rock-magnetic experiments and electron microprobe analyses indicate that the primary Fe–Ti oxides are the main magnetic carriers. All lava flows carry normal palaeomagnetic directions, which can be grouped into 31 units by an *F*-test, with a Fisher mean characteristic remanent magnetization (ChRM) of $D/I = 12.8/58.6^\circ$ ($\alpha_{95} = 2.3^\circ$). The corresponding palaeomagnetic pole is located at 80.3°N and 200.3°E ($A_{95} = 3.2^\circ$, $K = 64.4$), which is indistinguishable at the 95% confidence level from the Eurasia pole derived from the apparent polar wander path for the early Cretaceous. Using a modified Thellier palaeointensity method with stringent acceptance criteria, we obtained two time-series of palaeointensity records from 15 independent palaeomagnetic units (total 136 samples). The virtual dipole moment (VDM) values varied from 2.53×10^{22} Am^2 to 9.92×10^{22} Am^2. The mean VDMs for the upper and lower intervals are $(5.38 \pm 2.06) \times 10^{22}$ Am^2 and $(4.61 \pm 2.67) \times 10^{22}$ Am^2, respectively. The observed time-series of palaeointensity, together with the previously available data for the CNS, suggest that magnetic field strength during the CNS fluctuated significantly with time.

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1. Introduction

A major feature of the geomagnetic field in the Phanerozoic is that three polarity superchrons (the Cretaceous normal superchron, the Kiaman reverse superchron, and the Moyero reverse superchron), long intervals of tens of millions of years without reversals, are separated by even longer time intervals when reversals were relatively frequent (Courtillot and Olson, 2007). The determination of geomagnetic field strength during these superchrons is therefore essential for understanding the working state of the geodynamo. Palaeointensity studies on whole rock and some submarine basaltic glass (SBG) samples yielded rather low palaeointensity for the Cretaceous normal superchron (CNS, 83–120 Ma), about half or even lower values when compared with the present-day field (Sherwood et al., 1993; Juarez et al., 1998; Tanaka and Kono, 2002; Zhu et al., 2004b; Shi et al., 2005; Riisager et al., 2003; Granot et al., 2007). In contrast, Tarduno et al. (2001, 2002) demonstrated that the dipole moments recorded by single plagioclase crystals from the Arctic region (~95 Ma old) and the Rajmahal Traps (113–116 Ma old) have higher intensities of 12.7 ± 0.7 Am^2 and 12.5 ± 1.4 Am^2, respectively, much higher than that of the present-day field. Tauxe and Staudigel (2004) reported that the field intensities at ~92 Ma from SBG samples are equivalent to that of the present-day field. The lack and the highly unevenly distributed palaeointensity data make it difficult to reveal the actual geomagnetic significance (Valet, 2003).

The relationship between polarity interval length and field intensity has long been debated. From a compilation of palaeointensity data obtained from SBG samples (0–160 Ma), Tauxe (2006) recently proposed a positive relationship between polarity interval length and average palaeointensity. Zhu et al. (2001, 2003, 2004a) found that palaeointensity determined from lava flows indicates that prior to the CNS palaeointensity is low supporting this. A weak correlation between the length of a given polarity interval and the average palaeointensity in the relative palaeointensity data was also demonstrated by Tauxe and Hartl (1997) and Constable et al. (1998). If a positive correlation exists, the field strength during the

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CNS should be higher than those of pre-CNS and the Cenozoic. Additional palaeointensity data for the CNS will help to solve the above arguments and decipher the mechanisms of the dynamo working during the superchron.

In this paper, we report new integrated full vector palaeomagnetic results, and $^{40}Ar/^{39}Ar$ dating from a lava sequence of Early Cretaceous age from Suhongtu in Inner Mongolia, China.

2. Geology and sampling

The study area is geologically located in the southern margin of the Central Asian Orogenic Belt (CAOB), a belt separating the Siberian craton in the north from the Tarim block, Alashan block and the North China craton in the south (Fig. 1a). The Solonker suture zone indicates a final closure of the Palaeo-Asian ocean in the Permian (Şengör and Natal'in, 1996; Xiao et al., 2004). Mesozoic regional extensional tectonics is the striking feature in the southern part of CAOB, as indicated by occurrences of extensional basin formations, metamorphic core complexes, and granitic magmatism (e.g., Ritts et al., 2001; Wang et al., 2004; Wu et al., 2002; Zheng et al., 1991). Wu et al. (2002) proposed that the early Cretaceous A-type granites in northeastern and northern China were possibly associated with rifting in eastern China at this time, associated with the onset of palaeo-Pacific subduction.

Early Cretaceous strata are widely distributed in the western part of Inner Mongolia (Fig. 1b). Upwardly, Early Cretaceous strata are named the Bayangobi formation, the Suhongtu formation, and the Wulansuhai formation. The Bayangobi formation and Wulansuhai formation are dominantly grayish to brownish sandstones, siltstones and mudstones. The Suhongtu formation is mainly composed of intermediate-basic eruptives, andesites and basalts, with at some locations intercalations of grayish green mudtones and sandstones, bearing Early Cretaceous fossils bivale *Sphaerium jeholense*, ostracod *Cypridea* cf. *kansuensis*, and dinosaur *Psittacosaurus* sp. (Regional Geology of Nei Mongol (Inner Mongolia) Autonomous Region, 1982). No radiometric dating has been done on these rocks before the present study.

The sampling section is geographically located about ~80 km south of the Mongolia–China border, between the Badan Jaran Desert and Lang Shan Mountain (Fig. 1b), with altitudes from 880 m to 1060 m (Fig. 1c). Ren et al. (2004) carried out a preliminary palaeomagnetic study on lava flows from the Suhongtu area. Ren et al. found that primary remanence is preserved in these volcanic rocks. Stable characteristic remanence components could be isolated by thermal demagnetization above 300 °C. Their observed palaeomagnetic pole from seven grouped directions is in agreement with those for the Siberia Craton and the North China Craton. In the present study, we carried out integrated palaeomagnetism, palaeointensity and geochronology studies on a consecutive 51-lava-flow profile, which is well exposed along a valley (location 41.216–41.235°N, 104.074–104.126°E). Bedding attitudes gently (<10°), with lava flow thicknesses varying from 2 m to 7 m (Fig. 1d). The lower part of the sequence is composed of relatively thicker lava flows, with some rich in agate-filled vesicles. No interbedded sedimentary horizons were observed between lava flows in the entire profile.

Seven to 11 palaeomagnetic cores were taken from the denser base of each lava flow using a portable gasoline powered diamond drill during two field seasons in 2001 and 2006. The cores were orientated in situ using magnetic compass and sun compass. A total of 434 cores were collected. Fresh massive samples were also collected for radiometric dating.

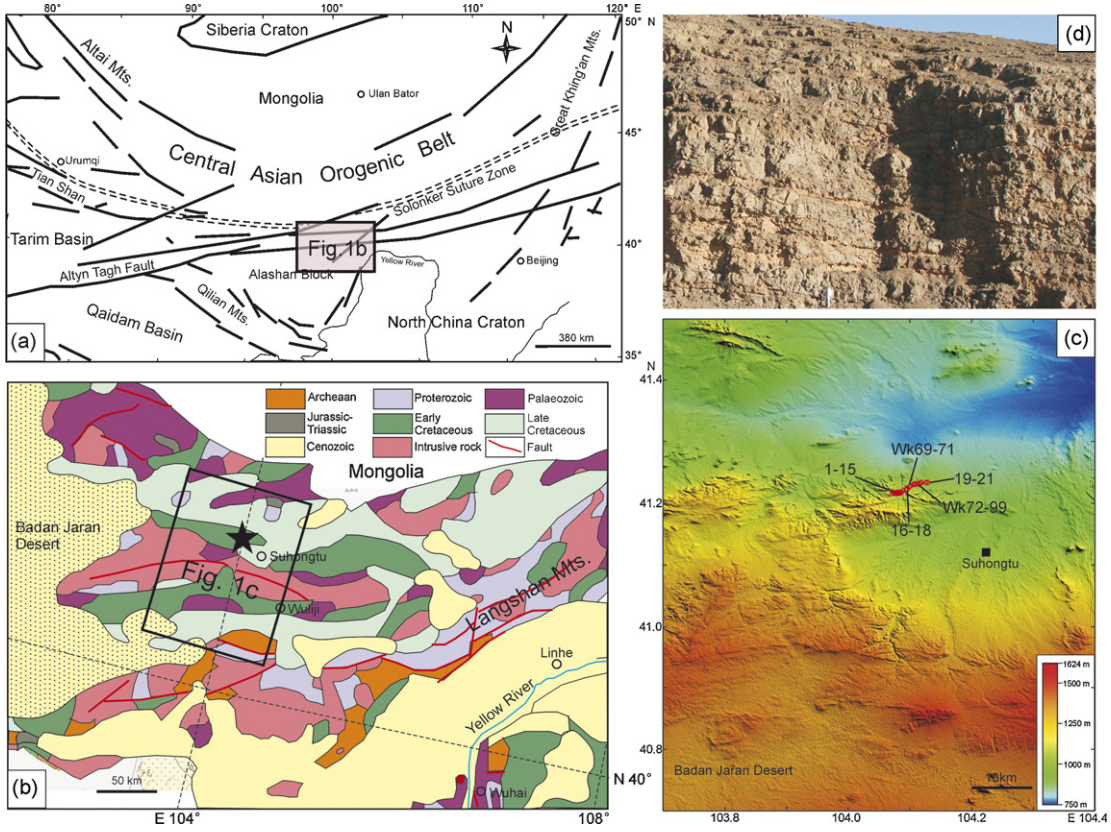


Fig. 1. Geologic sketch map showing the location of the Suhongtu section (a–c) and photos showing the outcrop of the sampling section (d).

Table 1
$^{40}Ar/^{39}Ar$ step-heating data for matrix from samples SH9, SH11, WK76, WK77, WK81, WK83, WK87, WK88, WK90,WK95, WK98, and WK99

Temp (°C)	$^{39}Ar/^{40}Ar \pm 1\sigma$ (10^{-2})	$^{36}Ar/^{40}Ar \pm 1\sigma$ (10^{-4})	$^{37}Ar/^{39}Ar$	^{39}Ar Cum (%)	$^{40}Ar^*$ (%)	$^{40}Ar^*/^{39}Ar$	Apparent Age $\pm 2\sigma$ (Ma)
Matrix separate from SH9 (weight = 3.72 mg, $J = 0.005295 \pm 0.000011$)							
820	6.26 ± 0.05	6.59 ± 0.13	0.47	5.47	80.52	12.87	119.20 ± 1.14
860	4.02 ± 0.05	17.13 ± 0.28	0.47	7.75	49.39	12.27	113.85 ± 2.33
900	7.18 ± 0.05	3.52 ± 0.07	0.47	15.45	89.59	12.49	115.76 ± 0.78
930	7.91 ± 0.05	0.75 ± 0.03	0.40	16.00	97.77	12.36	114.62 ± 0.74
960	7.92 ± 0.07	0.99 ± 0.03	0.38	12.67	97.08	12.26	113.74 ± 1.04
990	7.17 ± 0.04	4.29 ± 0.09	0.36	8.34	87.33	12.18	113.05 ± 0.76
1030	7.37 ± 0.07	3.31 ± 0.06	0.28	6.66	90.21	12.25	113.64 ± 1.05
1080	7.74 ± 0.06	2.38 ± 0.07	0.24	8.22	92.97	12.02	111.54 ± 0.91
1130	7.69 ± 0.05	2.34 ± 0.08	0.35	6.57	93.09	12.11	112.36 ± 0.79
1190	7.82 ± 0.08	2.36 ± 0.05	0.54	7.12	93.03	11.89	110.42 ± 1.06
1260	7.54 ± 0.11	3.98 ± 0.14	1.22	2.54	88.22	11.71	108.75 ± 1.58
1400	7.38 ± 0.10	4.41 ± 0.16	1.15	3.20	86.98	11.78	109.45 ± 1.61
Matrix separate from SH11 (weight = 3.6 mg, $J = 0.005318 \pm 0.000011$)							
760	4.96 ± 0.15	11.93 ± 0.22	0.46	2.08	64.76	13.06	121.17 ± 3.83
800	6.19 ± 0.17	7.81 ± 0.17	0.48	2.96	76.91	12.43	115.47 ± 3.23
840	3.74 ± 0.03	18.49 ± 0.32	0.48	6.35	45.35	12.11	112.60 ± 2.50
870	4.85 ± 0.05	14.09 ± 0.24	0.46	8.66	58.37	12.04	111.96 ± 1.81
900	7.71 ± 0.07	1.74 ± 0.04	0.41	11.66	94.85	12.31	114.37 ± 0.98
930	7.96 ± 0.05	0.83 ± 0.03	0.39	12.91	97.55	12.26	113.94 ± 0.72
960	7.94 ± 0.06	0.95 ± 0.03	0.36	11.92	97.20	12.25	113.81 ± 0.82
1000	7.79 ± 0.04	1.62 ± 0.03	0.35	11.89	95.21	12.23	113.64 ± 0.63
1050	7.74 ± 0.07	2.09 ± 0.21	0.32	10.11	93.82	12.12	112.65 ± 1.26
1110	7.34 ± 0.10	4.06 ± 0.31	0.33	6.81	87.99	11.99	111.50 ± 1.81
1180	7.35 ± 0.05	5.25 ± 0.23	0.47	9.69	84.47	11.49	106.96 ± 1.09
1270	7.06 ± 0.19	6.45 ± 0.79	1.17	2.50	80.93	11.47	106.81 ± 4.11
1400	6.88 ± 0.20	7.30 ± 0.78	1.20	2.47	78.42	11.40	106.18 ± 4.31
Matrix separate from WK76 (weight = 3.46 mg, $J = 0.005088 \pm 0.000025$)							
800	3.57 ± 0.07	18.34 ± 0.47	0.46	5.44	45.79	12.81	114.20 ± 4.75
850	1.59 ± 0.01	27.42 ± 0.51	0.35	11.15	18.97	11.97	106.88 ± 8.29
880	6.75 ± 0.06	5.14 ± 0.12	0.36	15.72	84.80	12.57	112.08 ± 1.12
970	7.47 ± 0.11	2.50 ± 0.08	0.37	3.59	92.61	12.39	110.55 ± 1.60
1010	6.45 ± 0.12	6.74 ± 0.14	0.38	4.10	80.07	12.42	110.81 ± 2.12
1050	7.31 ± 0.16	3.09 ± 0.09	0.39	4.70	90.87	12.42	110.85 ± 2.33
1100	7.29 ± 0.09	3.27 ± 0.07	0.42	13.86	90.35	12.39	110.59 ± 1.40
1150	7.32 ± 0.07	3.44 ± 0.08	0.58	14.26	89.83	12.28	109.58 ± 1.16
1220	7.37 ± 0.06	3.53 ± 0.08	2.07	22.11	89.56	12.15	108.49 ± 0.94
1300	6.95 ± 0.12	4.96 ± 0.14	6.52	4.21	85.35	12.29	109.68 ± 1.99
1400	5.43 ± 0.28	11.73 ± 0.51	9.12	0.84	65.32	12.03	107.44 ± 6.05
Matrix separate from WK77 (weight = 4.12 mg, $J = 0.009577 \pm 0.000048$)							
780	3.93 ± 0.11	16.39 ± 0.55	0.71	2.64	51.58	13.13	116.92 ± 6.00
830	1.26 ± 0.01	27.76 ± 0.54	0.60	6.38	17.96	14.27	126.73 ± 11.19
870	4.88 ± 0.07	13.24 ± 0.27	0.66	10.71	60.87	12.47	111.17 ± 2.16
910	7.61 ± 0.08	1.80 ± 0.05	0.60	13.61	94.68	12.44	110.96 ± 1.17
950	7.47 ± 0.07	2.38 ± 0.06	0.42	8.76	92.97	12.45	111.00 ± 1.11
990	6.92 ± 0.08	4.43 ± 0.10	0.44	6.78	86.90	12.56	111.97 ± 1.34
1030	6.01 ± 0.08	8.83 ± 0.20	0.55	5.66	73.90	12.30	109.71 ± 1.61
1080	7.07 ± 0.08	4.02 ± 0.12	0.62	7.01	88.10	12.46	111.14 ± 1.33
1130	7.35 ± 0.11	3.06 ± 0.08	0.63	10.33	90.95	12.38	110.44 ± 1.61
1180	7.39 ± 0.09	2.94 ± 0.08	0.95	12.67	91.31	12.35	110.19 ± 1.36
1240	7.28 ± 0.07	3.38 ± 0.09	4.02	11.41	90.02	12.36	110.25 ± 1.18
1400	6.49 ± 0.06	6.66 ± 0.18	11.56	4.04	80.31	12.37	110.37 ± 1.31
Matrix separate from WK81 (weight = 3.71 mg, $J = 0.005098 \pm 0.000026$)							
800	3.67 ± 0.06	18.96 ± 0.40	0.63	2.72	43.97	11.98	107.16 ± 3.69
860	1.24 ± 0.01	29.60 ± 0.55	0.46	6.17	12.53	10.07	90.56 ± 11.41
900	5.65 ± 0.05	10.35 ± 0.20	0.43	10.24	69.40	12.28	109.77 ± 1.41
940	7.53 ± 0.07	2.25 ± 0.05	0.39	14.63	93.35	12.39	110.79 ± 1.06
980	7.06 ± 0.07	4.39 ± 0.09	0.35	11.89	87.01	12.33	110.22 ± 1.14
1020	6.54 ± 0.07	6.77 ± 0.15	0.43	6.61	79.99	12.23	109.38 ± 1.33
1070	6.43 ± 0.08	7.38 ± 0.16	0.47	7.40	78.17	12.16	108.80 ± 1.53
1120	6.58 ± 0.06	6.97 ± 0.14	0.47	9.32	79.40	12.07	107.98 ± 1.23
1160	6.53 ± 0.06	7.47 ± 0.16	0.52	9.32	77.91	11.93	106.78 ± 1.30
1200	6.40 ± 0.08	7.88 ± 0.19	0.99	8.15	76.71	11.99	107.26 ± 1.65
1250	6.29 ± 0.05	8.26 ± 0.17	2.56	10.76	75.57	12.01	107.49 ± 1.19
1400	5.55 ± 0.09	11.44 ± 0.25	7.08	2.79	66.19	11.93	106.78 ± 2.00
Matrix separate from WK83 (weight = 3.79 mg, $J = 0.005105 \pm 0.000026$)							
800	2.00 ± 0.01	26.04 ± 0.48	0.25	8.11	23.05	11.51	103.23 ± 6.22
830	7.34 ± 0.08	2.69 ± 0.06	0.23	9.14	92.05	12.55	112.25 ± 1.32
860	7.73 ± 0.11	1.39 ± 0.03	0.25	9.17	95.90	12.41	111.09 ± 1.58
890	7.59 ± 0.12	1.82 ± 0.05	0.30	6.82	94.61	12.47	111.60 ± 1.74
930	7.16 ± 0.15	3.77 ± 0.11	0.38	5.55	88.86	12.40	111.03 ± 2.28

Table 1 (*Continued*)

Temp (°C)	$^{39}Ar/^{40}Ar \pm 1\sigma$ (10^{-2})	$^{36}Ar/^{40}Ar \pm 1\sigma$ (10^{-4})	$^{37}Ar/^{39}Ar$	^{39}Ar Cum (%)	$^{40}Ar^*$ (%)	$^{40}Ar^*/^{39}Ar$	Apparent Age $\pm 2\sigma$ (Ma)
980	6.97 ± 0.08	4.92 ± 0.10	0.47	6.53	85.47	12.26	109.75 ± 1.30
1030	7.27 ± 0.07	3.50 ± 0.08	0.52	9.72	89.66	12.33	110.39 ± 1.15
1070	7.56 ± 0.09	2.45 ± 0.06	0.60	10.56	92.74	12.27	109.86 ± 1.32
1110	7.70 ± 0.09	1.93 ± 0.05	0.72	8.83	94.30	12.24	109.63 ± 1.25
1150	7.68 ± 0.12	2.14 ± 0.06	0.95	9.15	93.68	12.20	109.23 ± 1.66
1200	7.61 ± 0.08	2.54 ± 0.07	2.60	12.07	92.49	12.15	108.83 ± 1.22
1250	7.22 ± 0.23	3.62 ± 0.16	7.66	2.27	89.29	12.37	110.72 ± 3.48
1400	6.77 ± 0.10	5.83 ± 0.16	9.43	2.08	82.76	12.23	109.52 ± 1.76
Matrix separate from WK87 (weight = 3.71 mg, $J = 0.005114 \pm 0.000026$)							
780	2.87 ± 0.09	20.82 ± 0.56	0.55	1.37	38.46	13.39	119.75 ± 7.02
850	0.71 ± 0.01	30.70 ± 0.58	0.36	4.86	9.28	13.02	116.59 ± 21.15
890	5.91 ± 0.08	9.00 ± 0.21	0.27	6.50	73.41	12.42	111.35 ± 1.97
930	7.51 ± 0.12	2.26 ± 0.06	0.31	6.20	93.32	12.43	111.43 ± 1.73
970	7.48 ± 0.08	2.68 ± 0.06	0.40	6.41	92.08	12.31	110.38 ± 1.26
1010	6.37 ± 0.11	7.17 ± 0.15	0.49	6.55	78.81	12.36	110.86 ± 2.01
1050	7.57 ± 0.09	2.13 ± 0.06	0.49	8.54	93.70	12.37	110.93 ± 1.30
1090	7.73 ± 0.06	1.61 ± 0.04	0.41	12.28	95.25	12.32	110.47 ± 0.91
1130	7.86 ± 0.08	1.23 ± 0.04	0.40	13.06	96.35	12.26	109.94 ± 1.14
1180	7.89 ± 0.11	0.97 ± 0.03	0.54	16.93	97.12	12.31	110.44 ± 1.45
1230	7.87 ± 0.08	1.06 ± 0.04	1.69	12.84	96.85	12.31	110.43 ± 1.16
1300	7.43 ± 0.15	3.07 ± 0.11	9.95	3.86	90.92	12.24	109.76 ± 2.22
1400	5.27 ± 0.47	12.41 ± 0.49	9.67	0.60	63.33	12.02	107.85 ± 9.63
Matrix separate from WK88 (weight = 3.70 mg, $J = 0.005126 \pm 0.000026$)							
800	4.40 ± 0.09	15.73 ± 0.41	0.44	3.59	53.51	12.17	109.42 ± 3.78
850	1.93 ± 0.02	26.78 ± 0.50	0.35	7.88	20.87	10.82	97.57 ± 6.76
890	5.22 ± 0.04	12.52 ± 0.24	0.36	11.31	63.01	12.08	108.63 ± 1.54
930	7.79 ± 0.06	1.33 ± 0.03	0.53	17.45	96.05	12.33	110.82 ± 0.88
970	7.64 ± 0.07	1.91 ± 0.05	0.55	10.67	94.34	12.35	111.02 ± 1.04
1010	7.20 ± 0.07	4.04 ± 0.09	0.56	7.48	88.06	12.23	109.96 ± 1.15
1050	7.04 ± 0.07	4.74 ± 0.11	0.51	7.12	85.99	12.21	109.79 ± 1.18
1100	7.54 ± 0.09	2.87 ± 0.08	0.45	8.53	91.50	12.14	109.18 ± 1.38
1150	7.63 ± 0.08	2.57 ± 0.06	0.53	9.51	92.39	12.11	108.90 ± 1.13
1200	7.21 ± 0.09	4.44 ± 0.11	1.55	6.63	86.86	12.05	108.37 ± 1.42
1270	7.35 ± 0.09	3.70 ± 0.10	4.54	7.12	89.05	12.12	109.00 ± 1.35
1400	5.59 ± 0.11	11.55 ± 0.25	6.72	2.70	65.86	11.78	106.06 ± 2.34
Matrix separate from WK90 (weight = 3.37 mg, $J = 0.005126 \pm 0.000026$)							
800	3.96 ± 0.16	17.31 ± 0.48	0.38	2.76	48.83	12.34	111.56 ± 5.73
840	3.35 ± 0.05	20.55 ± 0.43	0.24	4.72	39.27	11.73	106.15 ± 4.04
880	2.17 ± 0.01	25.85 ± 0.48	0.20	8.50	23.61	10.88	98.71 ± 5.83
920	7.40 ± 0.07	2.91 ± 0.06	0.19	13.45	91.39	12.35	111.59 ± 1.13
960	7.73 ± 0.08	1.63 ± 0.04	0.23	11.45	95.16	12.31	111.30 ± 1.10
1000	7.15 ± 0.05	4.30 ± 0.10	0.36	6.05	87.29	12.21	110.41 ± 0.84
1040	7.33 ± 0.08	3.42 ± 0.08	0.44	6.70	89.89	12.27	110.89 ± 1.30
1080	7.61 ± 0.09	2.45 ± 0.06	0.47	9.14	92.74	12.19	110.26 ± 1.25
1120	7.73 ± 0.08	2.03 ± 0.06	0.54	10.18	94.00	12.16	109.99 ± 1.13
1160	7.66 ± 0.08	2.55 ± 0.07	0.60	8.96	92.44	12.06	109.11 ± 1.12
1210	7.49 ± 0.09	3.13 ± 0.08	1.47	10.35	90.73	12.11	109.52 ± 1.40
1270	7.14 ± 0.12	4.55 ± 0.15	4.02	4.14	86.55	12.12	109.58 ± 1.89
1400	6.38 ± 0.07	7.96 ± 0.18	5.08	3.60	76.47	11.98	108.38 ± 1.36
Matrix separate from WK95 (weight = 3.77 mg, $J = 0.005174 \pm 0.000026$)							
870	3.99 ± 0.04	18.58 ± 0.35	0.20	3.23	45.09	11.29	102.67 ± 2.49
910	6.99 ± 0.07	5.21 ± 0.11	0.18	7.49	84.61	12.10	109.83 ± 1.25
950	7.61 ± 0.10	2.22 ± 0.07	0.20	8.25	93.43	12.28	111.37 ± 1.40
990	7.44 ± 0.10	3.36 ± 0.08	0.26	6.97	90.06	12.10	109.80 ± 1.45
1040	7.30 ± 0.07	3.73 ± 0.09	0.42	9.52	88.96	12.18	110.56 ± 1.09
1090	7.66 ± 0.06	2.31 ± 0.05	0.45	15.98	93.16	12.17	110.42 ± 0.86
1130	7.72 ± 0.06	2.14 ± 0.05	0.57	14.76	93.68	12.13	110.07 ± 0.88
1170	7.69 ± 0.09	2.31 ± 0.05	0.65	12.44	93.18	12.11	109.93 ± 1.24
1220	7.52 ± 0.06	3.14 ± 0.07	1.42	14.59	90.71	12.06	109.44 ± 0.85
1270	7.11 ± 0.08	4.82 ± 0.13	3.65	4.27	85.74	12.07	109.52 ± 1.39
1400	5.55 ± 0.08	12.18 ± 0.27	6.58	2.50	64.02	11.54	104.89 ± 2.09
Matrix separate from WK98 (weight = 3.39 mg, $J = 0.005170 \pm 0.000026$)							
800	4.96 ± 0.15	21.86 ± 0.58	0.44	1.95	35.40	11.76	106.73 ± 7.94
850	6.19 ± 0.01	31.11 ± 0.58	0.31	4.17	8.08	8.01	73.38 ± 15.21
900	3.74 ± 0.04	15.20 ± 0.31	0.25	6.68	55.07	12.00	108.82 ± 2.08
940	4.85 ± 0.08	4.07 ± 0.10	0.25	8.25	87.96	12.28	111.32 ± 1.27
980	7.71 ± 0.08	2.84 ± 0.06	0.30	9.32	91.61	12.27	111.24 ± 1.15
1020	7.96 ± 0.07	4.02 ± 0.09	0.37	10.28	88.10	12.20	110.61 ± 1.06
1060	7.94 ± 0.06	2.33 ± 0.06	0.41	10.39	93.12	12.25	111.07 ± 0.96
1100	7.79 ± 0.06	2.04 ± 0.06	0.40	11.25	93.96	12.28	111.33 ± 0.93
1150	7.74 ± 0.07	1.91 ± 0.04	0.42	14.95	94.36	12.19	110.53 ± 1.00

Table 1 (*Continued*)

Temp (°C)	$^{39}Ar/^{40}Ar \pm 1\sigma$ (10^{-2})	$^{36}Ar/^{40}Ar \pm 1\sigma$ (10^{-4})	$^{37}Ar/^{39}Ar$	^{39}Ar Cum (%)	$^{40}Ar^*$ (%)	$^{40}Ar^*/^{39}Ar$	Apparent Age $\pm 2\sigma$ (Ma)
1200	7.34 ± 0.06	3.15 ± 0.08	1.02	13.45	90.67	12.14	110.10 ± 1.00
1260	7.35 ± 0.07	3.87 ± 0.10	3.74	7.47	88.57	12.16	110.27 ± 1.08
1400	7.06 ± 0.11	12.96 ± 0.33	6.63	1.85	61.70	11.18	101.61 ± 2.47
Matrix separate from WK99 (weight = 3.36 mg, $J = 0.009370 \pm 0.000047$)							
700	6.94 ± 0.06	16.73 ± 0.61	0.61	0.68	50.57	7.28	119.38 ± 4.29
780	12.30 ± 0.06	2.85 ± 0.19	0.48	2.97	91.57	7.45	121.99 ± 0.95
860	10.20 ± 0.13	6.96 ± 0.19	0.22	15.28	79.42	7.78	127.29 ± 1.84
890	14.22 ± 0.07	0.53 ± 0.14	0.23	3.97	98.42	6.92	113.63 ± 0.74
920	14.20 ± 0.07	1.04 ± 0.07	0.25	7.52	96.93	6.83	112.14 ± 0.60
950	14.09 ± 0.07	1.34 ± 0.08	0.31	7.15	96.04	6.82	111.96 ± 0.60
980	14.03 ± 0.07	1.70 ± 0.09	0.39	7.59	94.97	6.77	111.25 ± 0.62
1010	13.53 ± 0.07	2.97 ± 0.10	0.46	8.10	91.21	6.74	110.82 ± 0.64
1040	13.79 ± 0.07	2.42 ± 0.08	0.52	8.65	92.84	6.73	110.64 ± 0.58
1070	13.79 ± 0.24	2.49 ± 0.08	0.67	9.63	92.62	6.72	110.40 ± 1.93
1100	13.62 ± 0.07	2.85 ± 0.09	0.72	8.04	91.57	6.72	110.46 ± 0.61
1130	14.05 ± 0.08	2.03 ± 0.08	0.76	7.15	93.99	6.69	109.96 ± 0.63
1180	14.36 ± 0.07	1.37 ± 0.08	1.34	6.53	95.95	6.68	109.81 ± 0.60
1240	14.33 ± 0.07	2.22 ± 0.11	3.14	5.06	93.43	6.52	107.27 ± 0.65
1300	20.63 ± 0.15	0.00 ± 0.60	4.16	1.34	136.79	6.63	109.04 ± 1.57
1500	7.02 ± 0.20	19.11 ± 1.17	0.96	0.34	43.53	6.20	102.15 ± 8.94

Note: All isotopes are corrected to Ca and K correction factors and decay time between the irradiation and measurements.

3. $^{40}Ar/^{39}Ar$ ages

Fresh massive samples were crushed and sieved between 40 mesh and 80 mesh (380–200 μm) and washed with distilled water. Volcanic matrixes without phenocrysts were hand picked under binocular microscope and washed with acetone in an ultra-sonic bath for 20 min. To remove possible alteration, matrix samples were washed with 5% HNO_3 in an ultra-sonic bath for 20 min. The grains were then rinsed with distilled water and dried, and were wrapped in Al foil and irradiated together with TCR-sanidine standards, optical CaF_2 and K-glass monitors in position H8 of the 49-2 reactor, Beijing, China, for 45 h (samples WK77 and WK99) or 23.5 h for other samples with 0.5 mm cadmium foil shield. To be concordant with GTS04, we use the reference age 28.34 ± 0.28 Ma (Renne et al., 1998) for TCR-sanidine standards. Ca, K correction factors were calculated from the CaF_2 and K-glass monitors: $(^{40}Ar/^{39}Ar)_K = 8.8 \times 10^{-4}$, $(^{39}Ar/^{37}Ar)_{Ca} = 7.92 \times 10^{-4}$, $(^{36}Ar/^{37}Ar)_{Ca} = 2.21 \times 10^{-4}$. The total fusion of standards and step-heating analysis of samples was performed at the Palaeomagnetism and Geochronology Laboratory (PGL), Institute of Geology and Geophysics (IGG), Chinese Academy of Sciences, on a MM5400 mass spectrometer operating in a static mode. The data were corrected for system blanks, mass discriminations, interfering argon isotopes derived from Ca and K and decay of ^{37}Ar since the time of irradiation. The decay constant used throughout the calculations is $\lambda = (5.543 \pm 0.010) \times 10^{-10}\ a^{-1}$, as recommended by Steiger and Jäger (1977). Detailed analytical and data processing procedures are outlined in He et al. (2004). The results of the $^{40}Ar/^{39}Ar$ step-heating are shown in Table 1 and plotted as age spectrum and inverse isochron in Fig. 2. As the isochron age makes no assumption regarding the trapped component, the isochron ages are preferred over the weighted mean plateau ages.

Lava flows SH9 and SH11 yield concordant age spectrums (Fig. 2A–D), with plateau segments comprising 52–83% of the gas released, inverse isochron ages are in good agreement with the plateau ages; and the $^{40}Ar/^{36}Ar$ intercept values are not distinguishable from the air ratio, indicating that there is no apparent excess argon contamination. Therefore, 114.9 ± 1.3 Ma (2σ) and 113.8 ± 0.7 Ma (2σ) are a good estimate of the eruption age of lava SH9 and SH11. These two ages are indistinguishable at the 95% confidence level, with the weighted mean age of 114.1 ± 0.3 Ma, representing an older eruption episode in the Suhongtu section.

Lava flows WK76, WK77, WK80, WK81, WK83, WK87, WK88, WK95, WK98, and WK99 also yield concordant age spectrums (Fig. 2E–X), with plateau segments comprising 49–94% of the gas released. Inverse isochron ages are in good agreement with the plateau ages; and the $^{40}Ar/^{36}Ar$ intercept values are not distinguishable from the air ratio at the 95% confidence level, indicating that there is no apparent excess argon contamination. The concordant age spectrum and the good agreement between plateau ages and isochron ages show no sign of recoil effect, which is often caused by alteration. Moreover, the good agreement among the inverse isochron ages of WK76, WK77, WK80, WK81, WK83, WK87, WK88, WK95, WK98, and WK99 indicates that the probability of alteration contamination is low. Thus the weighted mean age of 110.6 ± 0.1 Ma (2σ) represents a younger eruption episode in the Suhongtu section.

4. Petrologic analyses and rock-magnetic results

Light microscopic observations, electron microprobe analyses and rock-magnetic experiments were performed on representative samples to determine the magnetic mineralogy and thermal stability of the samples. Light microscopy (Zeiss Axioskop 40) observations on thin sections indicate that rocks are fresh porphyritic basalts and andesites with pilotaxitic textures in the matrix. The reaction rim texture and skeletal texture of phenocrystals (hornblende and ilmenite) suggest rapid cooling and high-temperature oxidation. Electron probe microanalyses (EPMA) on the coarse opaque minerals (>1 μm) were carried out on a JEOL JXA-8100 electron microprobe at the IGG. The EPMA analyses reveal Fe-oxide compositions (titanomagnetite, ilmenite-titanomagnetite intergrowth, and ulvöspinel). Ilmenite occurs as euhedral cubic, rhombohedra and rod grains (Fig. 3a–f). Some euhedral ilmenites are rimmed by thin corona of titanomagnetite (Fig. 3b, d and e), indicating that titanomagnetite is formed later than ilmenite. In cases, equilibrated intergrowth lammella of ilmenite and titanomagnetite are also observed (Fig. 3a and f).

Room temperature hysteresis loops were measured using a magnetic measurements variable field translation balance (MMVFTB). Representative hysteresis loops are shown in Fig. 4. The saturation remanent magnetization (SIRM) acquisition shows that samples were saturated (>90%) at a field of 0.3 T. The saturation remanent magnetization (M_{rs}), the saturation magnetization (M_s) and the coercive force (B_c) were calculated after removal of the para-

magnetic contribution. The coercivity of remanence (B_{cr}) was determined by applying progressively increasing backfield after saturation. B_c ranges from 3.7 to 37.6 mT, and B_{cr} from 15.6 to 73.9 mT. Ratios of hysteresis parameters (B_{cr}/B_c, 1.9–5.3; M_{rs}/M_s, 0.07–0.37) for the majority of samples fall in the pseudo-single-domain (PSD) range on a Day-plot (Day et al., 1977), or a mixture of single domain (SD) and multidomain (MD) grain sizes (Dunlop, 2002). The medium demagnetization field (MDF) determined by an alternating field demagnetization of the natural remanent magnetization (NRM) varies from 15 mT to 65 mT.

Temperature dependence of initial magnetic susceptibility (κ–T) curves was measured on 23 powder samples using a Kappa-

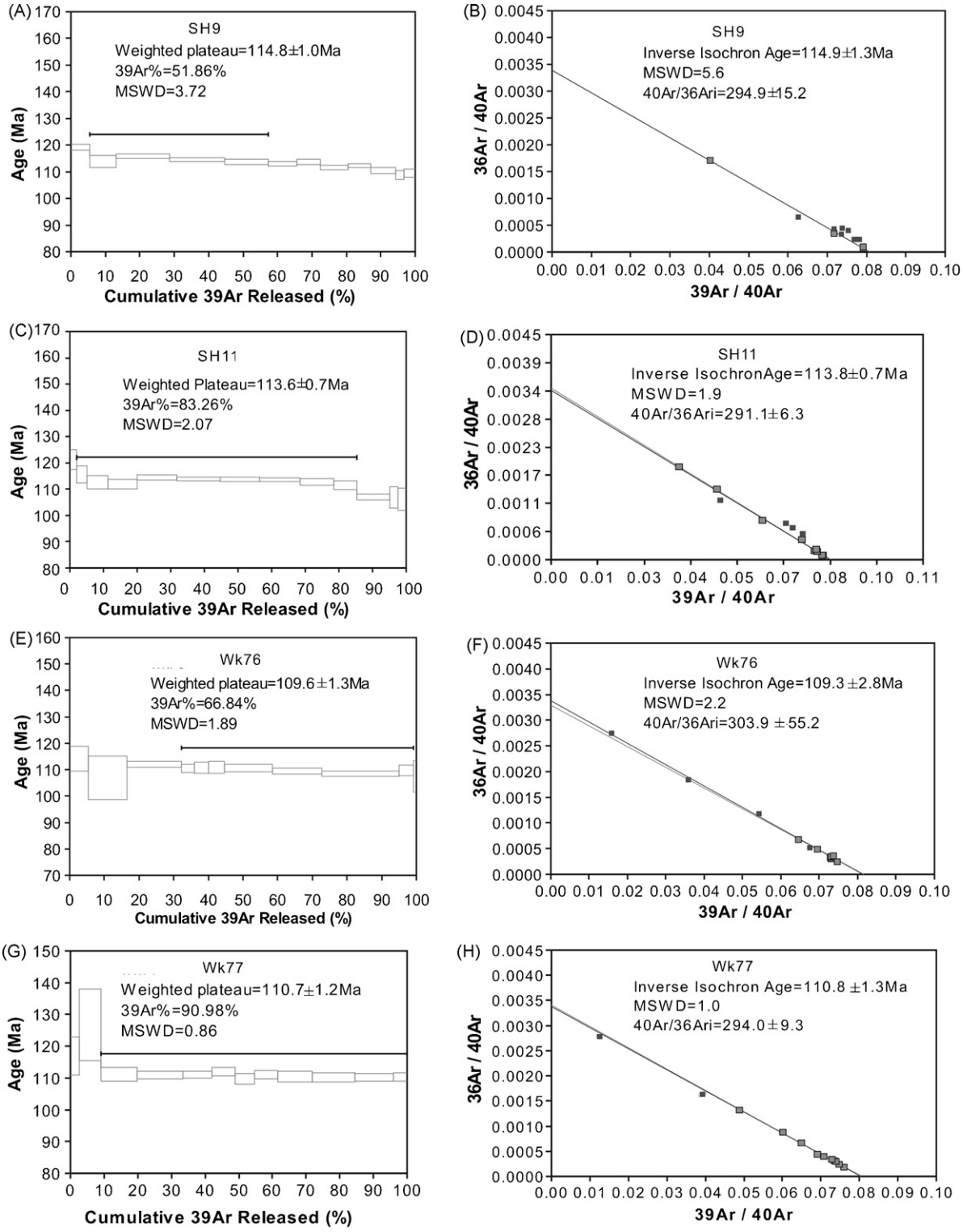


Fig. 2. Age spectra (left) and inverse isochronal plots (right) of matrix from fresh massive samples.

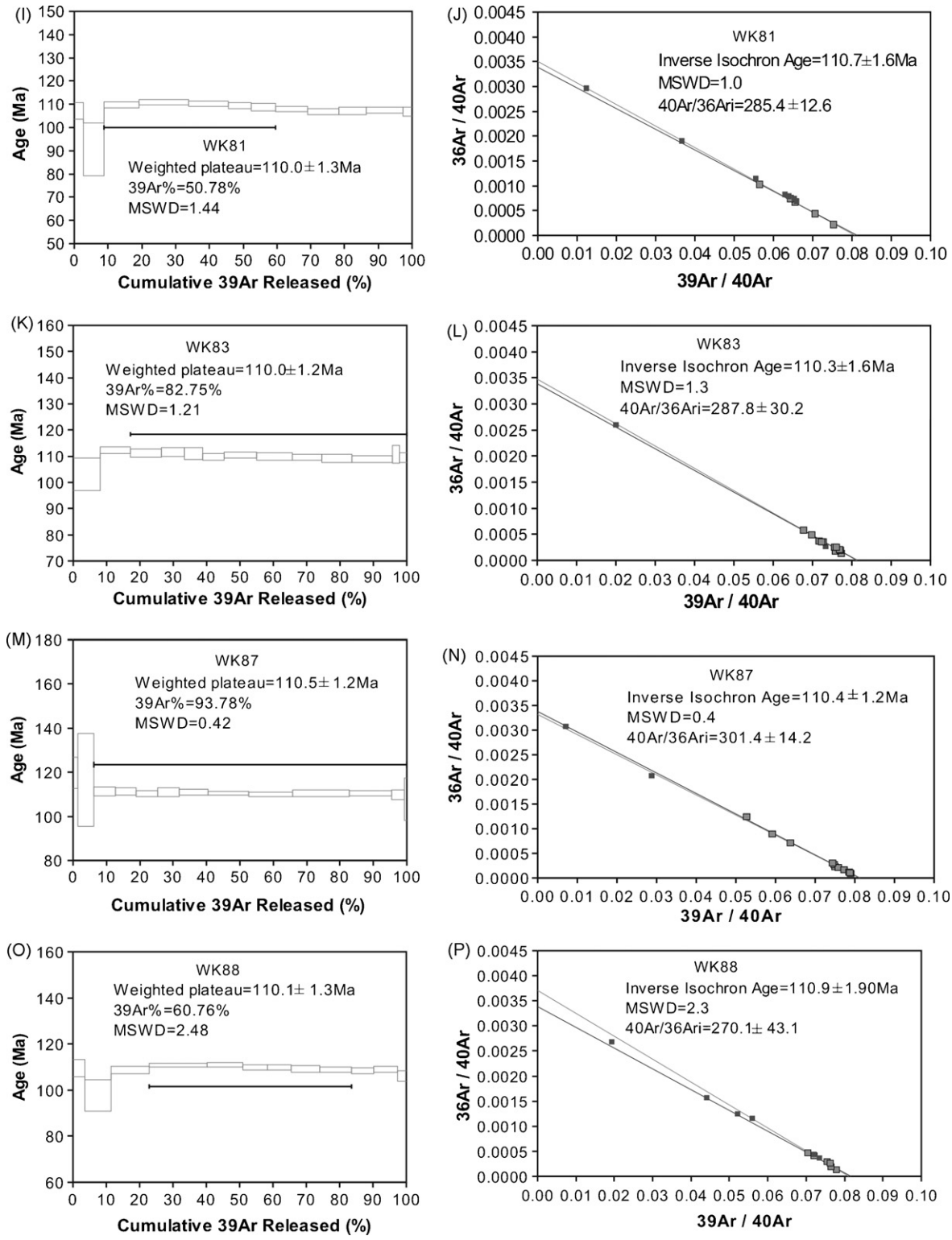


Fig. 2. (*Continued*)

bridge KLY-3 with attached temperature apparatus CS-2. The heating from room temperature up to 600 °C and cooling down back to room temperature with a rate about 10 °C min^{-1} were performed under argon atmosphere. Two magnetic phases can be recognized from the heating curves. On heating, the susceptibility of the first phase gradually increases up to 300 °C, followed by a drop to 450 °C. This phase disappears in most cooling curves, possibly suggesting an inversion of titanomagnetite or titanomaghemite (Vahle et al., 2007). The susceptibility of the second phase drops around 585 °C, indicating Ti-poor titanomagnetite. Representative κ–T curves are shown in Fig. 5. Further, more detailed rock-magnetic analyses of samples with the prefix WK (predominantly from the upper interval) can be found in Hill et al. (2008).

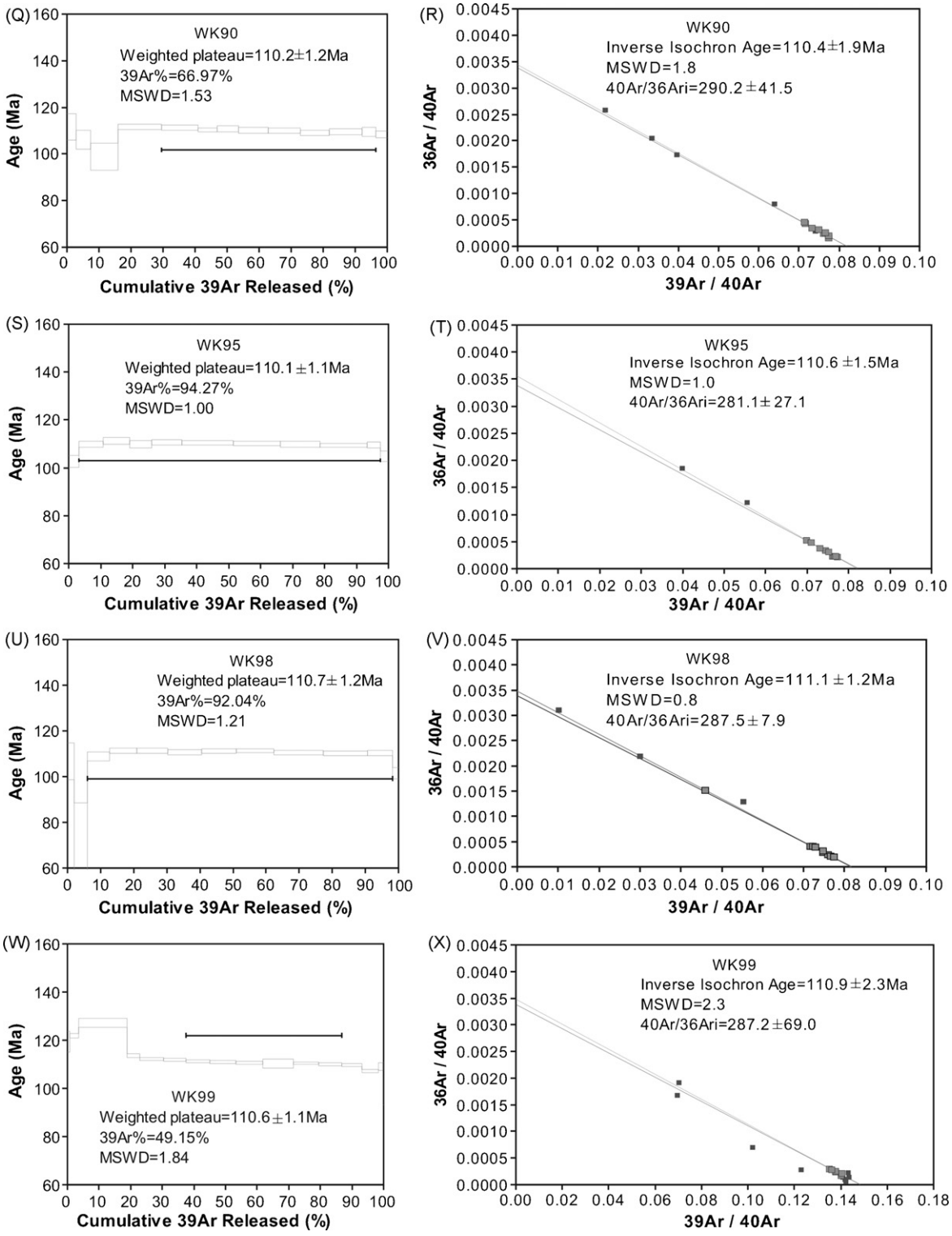


Fig. 2. (*Continued*).

5. Palaeodirectional results

The cylindrical cores with a diameter of 2.5 cm were cut to short cylinders with length of 1.0 cm in the laboratory. From the short specimens, a cylinder with diameter 1.0 cm was drilled using a stationary drill. The small cylindrical sample was measured for palaeointensity and the hollow ring used for stepwise thermal demagnetization. To isolate the characteristic remanent magnetization (ChRM) components, 434 hollow ring samples (one from each core) were stepwise thermally demagnetized in 12–15 steps from room temperature up to 680 °C using an ASC TD-48 thermal demagnetizer with residual field less than 10 nT and measured with a 2 G cryogenic magnetometer. All the instruments are situated in a field free space (<300 nT) of the PGL at the IGG. Examples

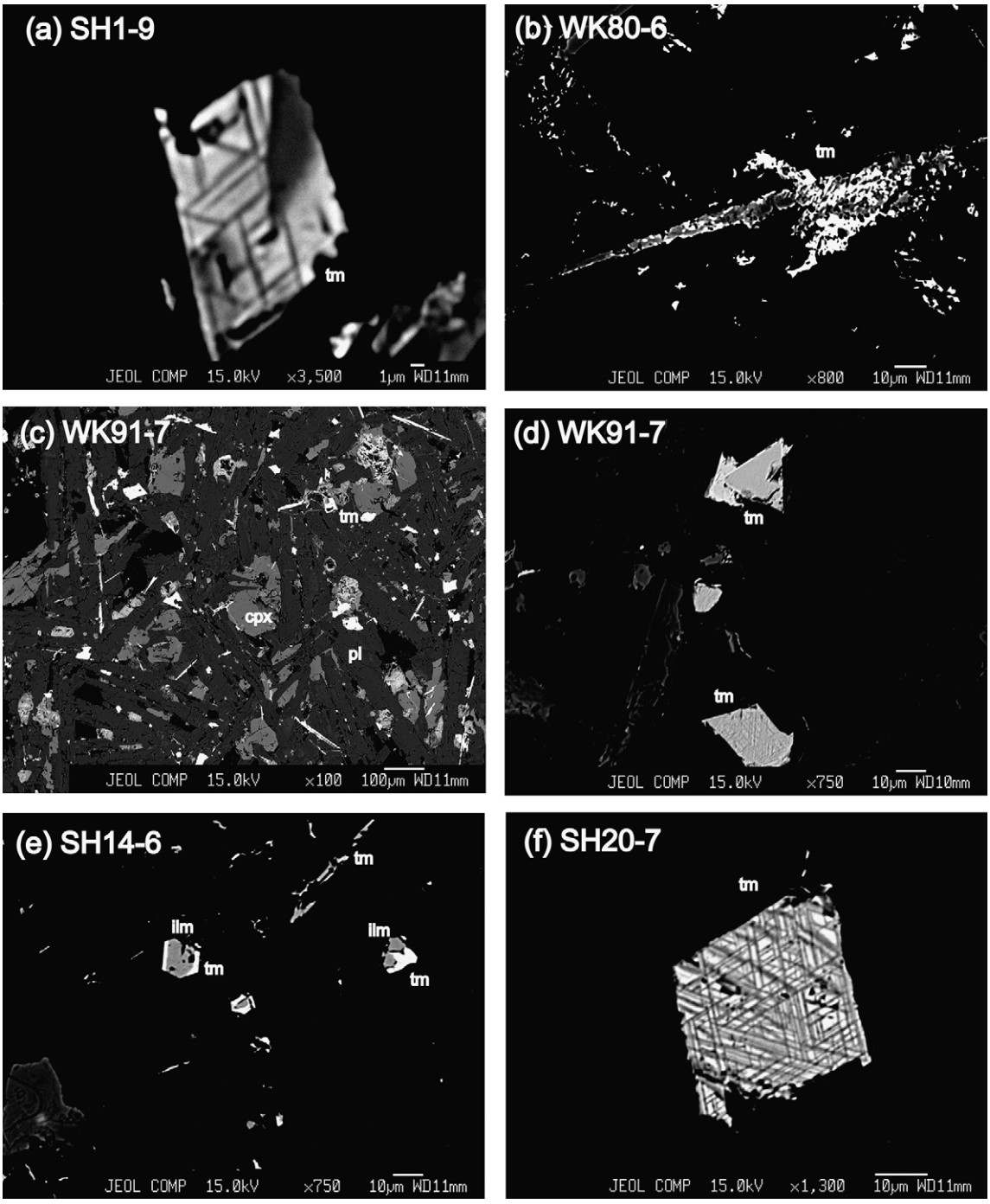


Fig. 3. Micrographs (images in backscatter mode) showing the magnetic minerals, titanomagnetite (tm), tm-ilmenite (ilm) intergrowths in the Suhongtu samples. (a) and (f) The equilibrated intergrowth lamellae of tm-ilm grains. (c) Opaque minerals (bright) filled in the plagioclase (pl) and pyroxene (cpx). (b), (d) and (e) Ilm rimed by tm.

of orthogonal projection plots are presented in Fig. 6. A secondary component, likely of viscous origin and generally small, was present in most samples and easily removed following thermal demagnetization to peak temperatures lower than 250–300 °C. Stable ChRM were revealed from 417 samples, and determined by fitting a straight line through at least five data points (Kirschvink, 1980) using the PMag software. Using Fisher statistics (Fisher, 1953), the mean direction for each lava was then determined. All lava flows are normally magnetized.

As the samples were collected from successive lava flows in a vertical sequence, a bias in the statistical analysis may be introduced by giving too much weight to a single direction recorded by successive lava flows emplaced in a short time interval. In order to probe whether these 51 lava flows recorded independent geomagnetic field directions palaeodirections from successive flows were compared. The probability of independence between successive lava flows using the *F*-test (e.g., Butler, 1992) was calculated. For values less than 95%, lava flows were assumed to have recorded the same geomagnetic field and were subsequently combined in to directional groups (DGs). However if the palaeointensity determined from one lava flow is obviously different from the other lava flows of the same directional group, we separated the lava from the

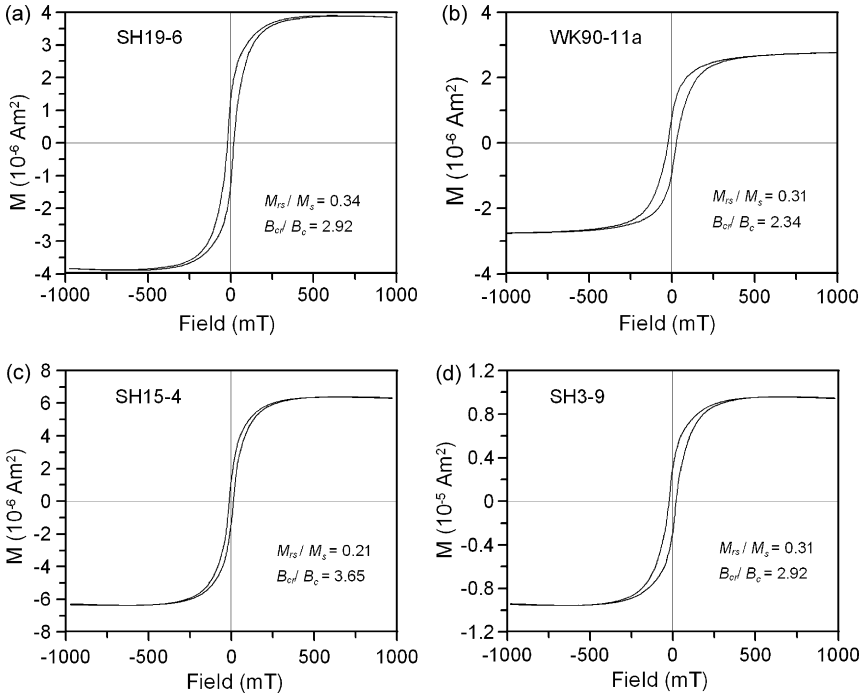


Fig. 4. Representative hysteresis loops measured at room temperature.

lava flow(s) below it in the group and redid the *F*-test (e.g., SH01 and SH20). Thirty-one DGs were obtained by this test. To calculate the DG mean direction all samples were given equal weight. The mean directions for each palaeomagnetically independent unit are shown in Table 2.

Taking the $^{40}Ar/^{39}Ar$ ages and palaeodirections, the entire sequence can be divided into the upper interval (WK71 to SH20) with weighted mean age of 110.6 ± 0.1 Ma and the lower interval (SH1 to WK70) with a weighted mean age of 114.1 ± 0.3 Ma. The mean directions are $D/I = 21.7^\circ/56.7^\circ$ ($\alpha_{95} = 2.0^\circ$, $N = 17$) and $D/I = 1.0^\circ/59.9^\circ$ ($\alpha_{95} = 2.2^\circ$, $N = 14$) for the upper and lower intervals, respectively. The overall mean direction is $D/I = 12.8^\circ/58.6^\circ$ ($\alpha_{95} = 2.3^\circ$, $N = 31$), corresponding to a palaeomagnetic pole for the sampling site with coordinates 80.3°N and 200.3°E ($A_{95} = 3.2^\circ$, $K = 64.4$). This is indistinguishable from the Eurasia pole derived from the APWP at the same time (110 Ma, 80.0°N, 183.6°E, $A_{95} = 4.2^\circ$, $K = 74.8$) (Besse and Courtillot, 2002). This consistence implies that (1) the primary magnetization direction has been preserved, (2) secular variation may have been averaged. We also note that there is a clear declination difference between directions of the upper and lower interval. The directional difference might be derived from palaeosecular variation, or, a 20.7° clockwise rotation of the upper interval respect to the lower interval. We prefer to interpret it as field behavior, because (1) no such vertical-axis rotation has been reported in this region; (2) the mean VGP pole for the entire section is indistinguishable from the expected value of the Eurasia pole.

The angular dispersion (*S*) was calculated after McFadden et al. (1991) for all 31 VGPs. *S* values of the upper interval and lower interval relative to their mean VGPs are 5.7 ($^{+1.5}/_{-1.5}$) and 6.2 ($^{+1.0}/_{-1.7}$). These very small *S* values suggest rapid (short time-span) eruptions, or, that the dipole was more dominant than the non-dipole components during the Cretaceous superchron relative to Brunhes chron. For the entire section, the *S* value relative to its mean VGP and the Eurasian pole of 110 Ma are 10.1 ($^{+1.6}/_{-1.2}$) and 10.5 ($^{+1.3}/_{-1.3}$), respectively, lower than the expected value from secular variation model G and for 110–80 Ma data (McFadden et al., 1991). For discussion of the palaeosecular variation during the CNS see Biggin et al. (2008).

To analyze further the nature of remanence, we separated the thermoremanences into the initial RM_{300} (NRM after demagnetization at 300 °C, at this point viscous remanent magnetization was removed), the middle-high temperature component (MHTC, below 585 °C, dominantly carried by titanomagnetite), high-temperature component (HTC, 585–630 °C) and highest temperature compo-

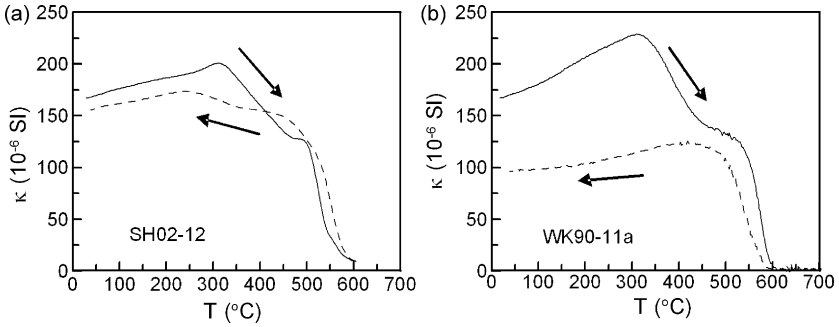


Fig. 5. Temperature dependence of initial magnetic susceptibility curves (κ–T) measured in argon atmosphere.

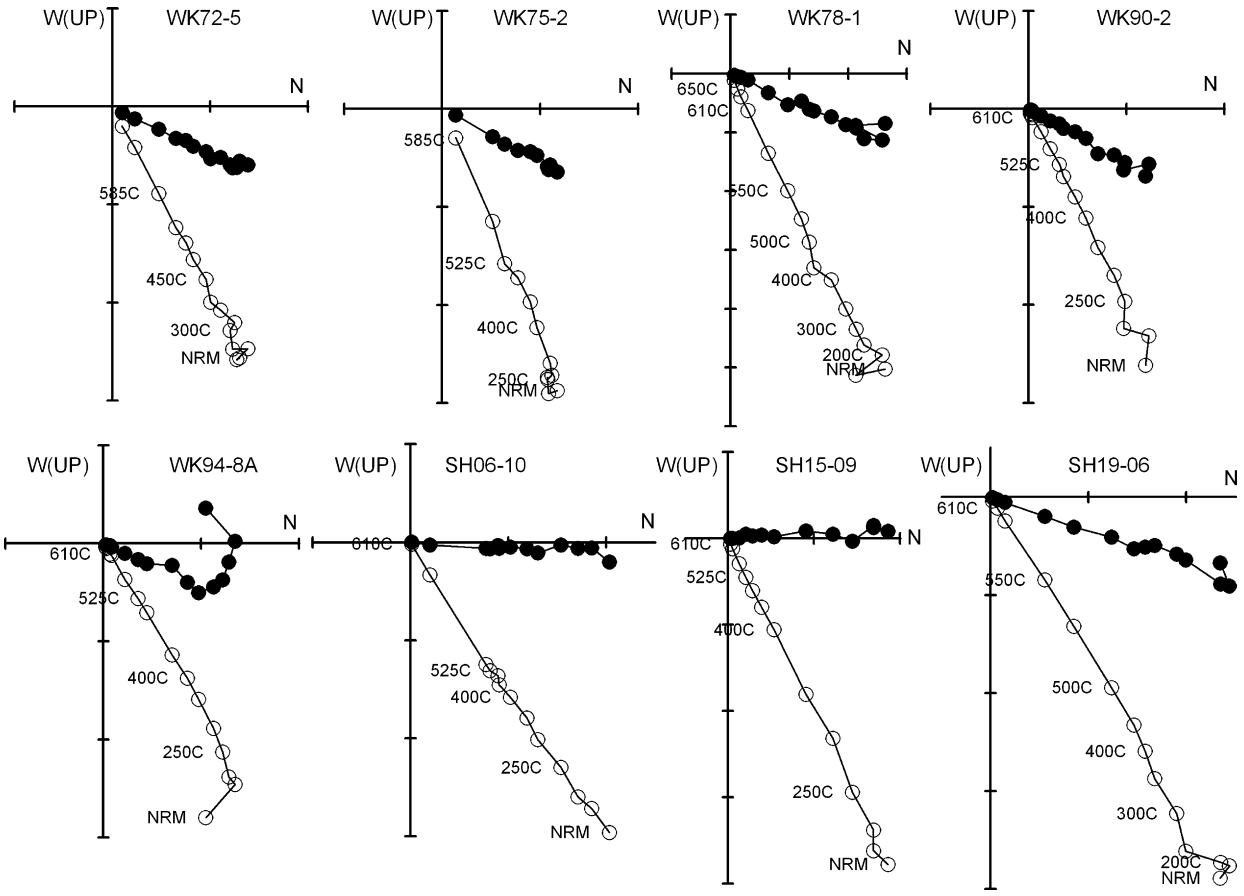


Fig. 6. Orthogonal projections of representative progressive thermal demagnetization. The solid (open) symbols refer to the horizontal (vertical) plane.

nent (HHTC, >630 °C). It has been reported that maghaemite has a higher Curie temperature than the inversion temperature (Dunlop and Özdemir, 1997). The HTC and HHTC components may be carried by titanomaghaemite and ilmenohaematite. 25 samples have HHTC components with a mean direction of $D/I = 1.2°/60.6°$ ($\alpha_{95} = 8.5°$). 376 samples have HTC components with a mean direction of $D/I = 12.5°/58.9°$ ($\alpha_{95} = 2.5°$). 417 samples have MHTC components with a mean direction of $D/I = 13.5°/58.7°$ ($\alpha_{95} = 2.3°$). This observation suggests that there is nearly no direction difference between the MHTC and HTC.

6. Palaeointensity determinations

Based on thermal demagnetization analyses, 23 samples were discarded for Thellier palaeointensity experiments due to relatively strong secondary overprinting. 411 samples were used for palaeointensity experiments. Prior to palaeointensity experiments the samples were stored in a shielded room to reduce possible viscous overprints. The modified version (Coe, 1967) of the Thellier and Thellier (1959) palaeointensity method was used in this study. A laboratory field intensity of 30 μT (or 60 μT) with a precision of 0.1 μT applied parallel to the *z*-axis of samples during both the heating and cooling cycles was used. A few milligrams of activated charcoal was introduced into the furnace prior to each run to act as a buffer to any oxidation of magnetic minerals (Laj and Kissel, 1999). Before heating, the furnace was pumped out and filled with argon gas, then pumped out again and refilled with argon again. Subsequently, a very small flow of argon gas was circulated during all heating and cooling cycles. At each temperature step, the samples were heated and cooled twice: in zero field for the first heating and under an applied field for the second heating. Above 200 °C, pTRM checks at a minimum of every other step were performed in order to detect any possible magnetomineralogical changes occurring during heating. Temperature reproducibility between two steps was in general better than 2 °C. The highest temperature used in the experiment was 610 °C with typical temperature intervals of 25–50 °C. pTRM tail checks were performed on samples with prefix SH.

Palaeointensities were calculated using a least-squares fit to a straight line through the selected temperature intervals. The NRM remaining at each temperature is plotted against the corresponding pTRM acquired in the laboratory field at the same temperature. Preference was given to the use of the higher temperature range and as large an NRM fraction as possible to calculate the palaeointensity (Chauvin et al., 2005). We imposed a stringent quality requirement: (i) at least five NRM–TRM points used to calculate the determination, and with the NRM fraction $f \geq 0.5$, to avoid possible bias of results by the presence of mutidomain grains (e.g., Biggin et al., 2007; Coe et al., 2004; Biggin and Thomas, 2003); (ii) the NRM directions obtained from the palaeointensity experiments head toward the origin with MAD <5° (Pick and Tauxe, 1993); (iii) the quality factor q is larger than 4 (except sample sh18–10 with $q = 3.9$) and the standard error of the slope divided by the slope is ≤0.1 (Coe et al. (1978); (iv) pTRM check (and tail check where present) less than 10% when normalized to total TRM. Among the 411 samples for palaeointensity experiments, 136 samples from 15 independent palaeomagnetic units met all the criteria discussed above and we interpret as yielding acceptable palaeointensity results (see Table 2; for individual sample data see online supplementary information). All samples which yielded linear behavior showed good internal

Table 2
Palaeomagnetic directions and palaeointensities determined from the Suhongtu section[a]

Interval	DG	Lava flows	*N*1	*Ds*	*Is*	*k*	α_{95}	PLA	PLO	*n*/*N*2	Ha	Std	VDM	Std
Upper interval	1	SH20	9	18.3	56.6	357.3	2.6	75.3	204.1	3/9	55.54	3.04	9.92	0.54
(~110 Ma)	2	SH19, WK97–98–99	28	14.5	57.6	114.9	2.6	78.5	204.4	12/21	28.25	1.42	4.98	0.25
	3	WK96	6	34.3	55.5	171.1	5.1	62.9	193.8					
	4	WK95	7	17.8	51.4	147.6	5	73.1	221.2					
	5	WK91–92–93–94	37	23.4	59.2	432.2	1.1	72.3	190.4	18/38	26.46	1.08	4.57	0.19
	6	WK89–90	24	20.6	56.5	137.1	2.5	73.5	202	12/21	17.69	0.84	3.16	0.15
	7	WK86–87–88	12	7.8	58.1	88.8	4.6	83.5	213.9					
	8	WK83–84–85	28	26	53.3	316	1.5	68.2	205.4	24/28	20.40	1.13	3.79	0.21
	9	WK82	10	23.4	57.9	350	2.6	71.9	194.9	5/10	21.20	1.83	3.73	0.32
	10	WK81	8	21.2	53.9	1511.5	1.4	72	209.4	5/8	30.08	1.44	5.56	0.26
	11	WK79–80–80b	29	26.9	55.2	178.4	2	68.3	200	12/25	35.51	2.38	6.46	0.43
	12	WK77-78	13	26.4	58.4	198.4	3	69.8	191.4	7/11	20.66	1.13	3.61	0.2
	13	WK76	6	23.3	50.9	328.5	3.7	69	213.9	3/7	40.23	3.74	7.70	0.71
	14	WK75-74	15	20.5	57.9	239.1	2.5	74.1	197	9/12	21.93	1.93	3.85	0.34
	15	WK73	3	18.2	64.2	1056.4	3.8	76	168.17	2/3	30.62	5.18	4.96	0.84
	16	WK72	7	26.2	58	622.5	2.4	69.8	192.7	4/6	40.93	1.66	7.18	0.29
	17	WK71	6	18.5	57.9	357.8	3.5	75.6	198.7					
	Mean			21.7	56.7	324.4	2.0	72.9	199.7		29.91	11.26	5.38	2.06
Lower interval	18	WK70	6	354.6	67.1	207.1	4.7	80.6	82.2					
(~114 Ma)	19	WK69	6	353.8	55.4	830.6	2.3	82.8	328.5					
	20	SH18	10	354.2	61.4	3520.8	0.8	85.5	32.7	8/10	15.08	0.49	2.53	0.08
	21	SH17	11	5.1	57.1	216.5	3.1	84.7	234.3					
	22	SH14–15–16	24	357.8	61.3	1200.8	0.9	88	50.5	3/18	45.33	4.02	7.62	0.68
	23	SH13	9	358.5	55	735.7	1.9	84.2	296.2					
	24	SH12	8	355.1	60.5	2229.4	1.2	86.3	19.4					
	25	SH10–11	19	359.2	63	694.3	1.3	86.7	94.1					
	26	SH09	10	5.8	59.4	793.1	1.7	85.5	205.2					
	27	SH08	8	0.2	56	318.8	3.1	85.3	282.1					
	28	SH06–07	16	357.4	59.6	1315.7	1	87.9	353.1					
	29	SH05	11	6.4	59	574.7	1.9	84.9	208.7					
	30	SH02–03–04	30	12.7	61.3	471.7	1.2	80.5	182.8	9/18	21.88	0.50	3.68	0.08
	31	SH01	8	12.6	60.8	368.8	2.7	80.6	186.4					
	Mean			1.0	59.9	314.2	2.2	89.2	214.4		27.43	15.87	4.61	2.67
	Section mean			12.8	58.6	125.6	2.3	80.3	200.3		29.41	11.69	5.22	2.11

[a] *N*1, number of samples used for the palaeodirection calculation; *Ds*/*Is*, declination/inclination of ChRMs; k/α_{95}, precise parameter/radius of circle of the 95% confidence about the direction; PLA/PLO, virtual geomagnetic pole (VGP) latitude/longitude. The latitude/longitude of the site is N/E; *n*/*N*2, the numbers of samples passing the palaeointensity quality criteria and samples used for the palaeointensity experiments; Ha and std, ancient field intensity (μT) and its standard deviation; VDM and std, the calculated virtual dipole moment and its standard deviation. Note that palaeointensity data of the lava flow WK73 are not used for the average due to its *n* less than 3.

consistency with low within-flow scatter. No reliable palaeointensity data were obtained from 16 palaeomagnetic units. Possible causes for the failure are discussed below.

Examples of the palaeointensity results, represented as NRM–TRM diagrams (Nagata et al., 1963) are shown in Fig. 7. We note that the samples yielding linear Arai diagrams were also characterized by high directional stability, with no tendency for the magnetization to align along the laboratory field (Laj and Kissel, 1999; Carlut and Quidelleur, 2000). The weighted mean palaeointensity value calculated for the palaeomagnetically independent lava flows (DG) vary from 15.08 μT to 55.54 μT, and the corresponding virtual dipole moment (VDM) values ranges from 2.53×10^{22} Am^2 to 9.92×10^{22} Am^2 (see Table 2), respectively.

As seen in Table 2 the mean VDMs for the upper and lower intervals are $(5.38 \pm 2.06) \times 10^{22}$ Am^2 and $(4.61 \pm 2.67) \times 10^{22}$ Am^2, respectively. Lava WK73 has only two palaeointensity estimates and thus was not included in the mean calculation. As a whole, all 15 palaeomagnetic independent units yield an overall mean VDM of $(5.22 \pm 2.11) \times 10^{22}$ Am^2. It is noted that the within-lava scatter of VDM of this study is very limited (standard deviation, std <10% of the mean). However, the overall-profile VDM variation (std) reaches as high as 40%. A microwave palaeointensity study on a selection of sister samples (all prefix WK and predominantly from the upper interval) has been carried out by Hill et al. in order to test the reproducibility of palaeointensity results (Hill et al., 2008) and they find similar variability.

7. Discussion

The ^{40}Ar/^{39}Ar ages and palaeomagnetic data show that the basaltic-andesitic lava sequence at the Suhongtu section emplaced during the CNS. The lower interval (SH01-WK70) was formed around ~114 Ma and the upper interval (WK71-SH20) around ~110 Ma (see Table 1). Within each of the two time intervals it is hard to distinguish lava flows by date indicating the flows were extruded in a short time interval. Rapid eruption is also supported by the monotonous petrologic nature, and absence of inter-bedded sediment layers. Stepwise thermal demagnetization reveals that ChRMs from all the studied lava flows have normal polarity (see Table 2). Light microscopy observations, EPMA and rock-magnetic measurements have indicated that Ti-poor titanomagnetite is the main remanence carriers of the studied samples. The textures of these oxides imply very rapid high-temperature oxidation. The consistence between the mean VGP of the present study and the corresponding Early Cretaceous Eurasian palaeomagnetic pole strongly suggests that primary magnetization directions have been preserved, secular variation might have been averaged, and the sampling site has no significant tectonic movements since the lava emplacement.

Maghaemite is problematic in palaeointensity determinations (e.g., Tarduno et al., 2006 and references therein), because low-temperature oxidation in nature results in maghemitization and a fundamental change in the nature of the magnetization, and struc-

turally maghaemite inverts to weakly magnetic hematite when heated in vacuum or in air by restacking of atomic planes (Dunlop and Özdemir, 1997; O'Reilly, 1983). We note that lavas SH5–13, SH17, WK70, WK71, WK86–88, WK95, WK96 have major intensity decay at temperatures of 300–450 °C and did not yield acceptable palaeointensity estimates. We interpret that maghaemite might be remanence carriers in these samples. For further discussion on the presence and effect of maghaemite in samples with prefix WK see Hill et al. (2008). The inversion temperatures of maghaemite may vary from 250 °C to ≥750 °C (Dunlop and Özdemir, 1997 and references therein). Maghaemite can be stable at 600 °C in an oxidizing environment for several hours, little maghaemite is present after heating for 2 h in an inactive atmosphere, while the maghaemite converted to magnetite after 0.5 h under reducing conditions (Helgason et al., 1992; Gunnlaugsson et al., 2002; Vahle et al., 2007). If inversion of maghaemite to hematite occurred by heating, the magnetic properties and remanence acquisition capability of the mother rocks must correspondingly change due to the magnetic differences between those phases. The room temperature coercivity and MDF values measured on heated samples do not increase after heating in argon to temperatures of 350 °C and 450 °C, compared with initial values of the unheated sample. Additionally, we would expect that the presence of maghaemite would be detected in the palaeointensity determination, but there is good linearity in the NRM–TRM plots (Fig. 7) and the magnetization heads toward the origin in the zero-field demagnetization steps of the Thellier palaeointensity experiments for the same temperature range. Therefore, we believe there is no clear evidence for an inversion of titanomaghaemite to hematite or magnetite in our palaeointensity determinations.

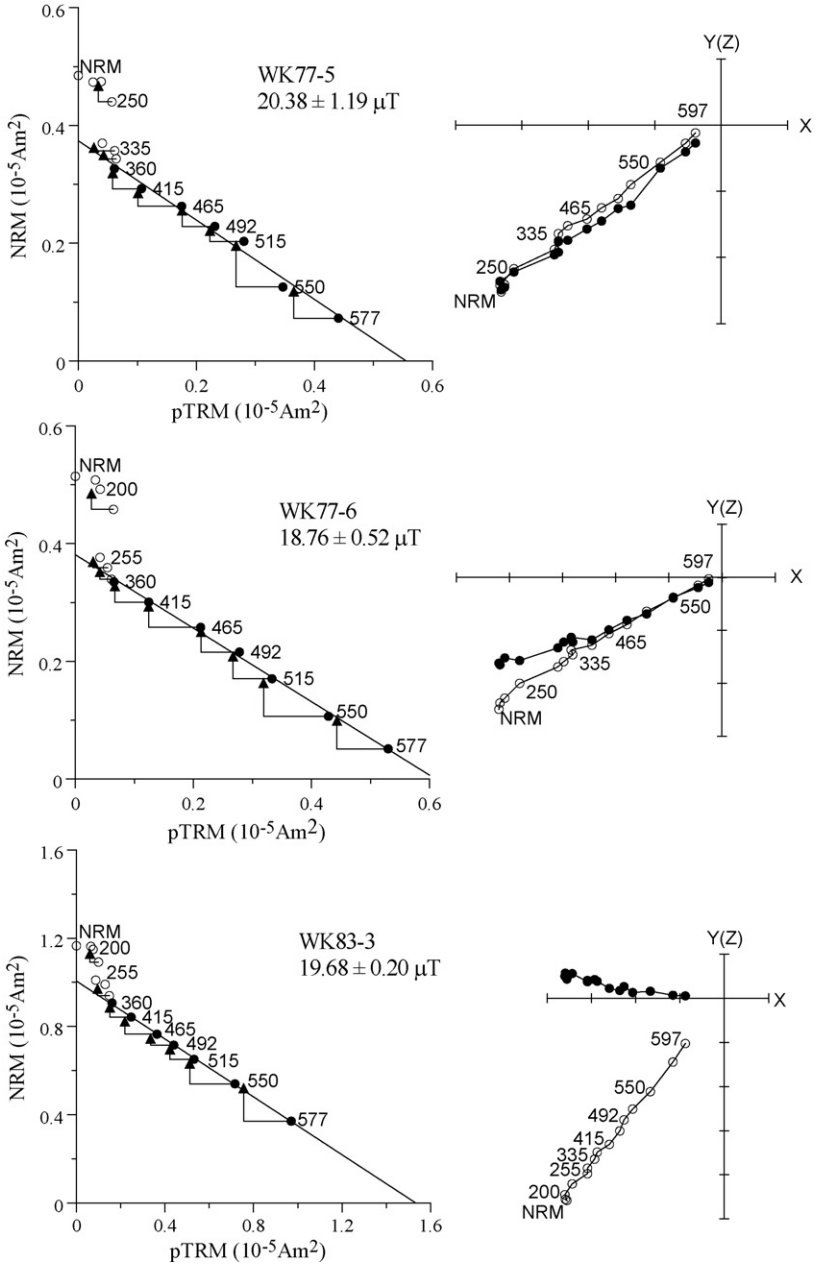


Fig. 7. Example of NRM–TRM plots (left) and associated thermal demagnetization behavior (right). Left, filled (open) circle denote data points accepted (rejected) from the best-fit line used to define the palaeointensity value, solid triangles are pTRM checks. Open triangles in plots of samples SH19–9 are pTRM tail checks. Right, orthogonal projection in sample coordinates, solid (open) symbols refer to the horizontal (vertical) plane.

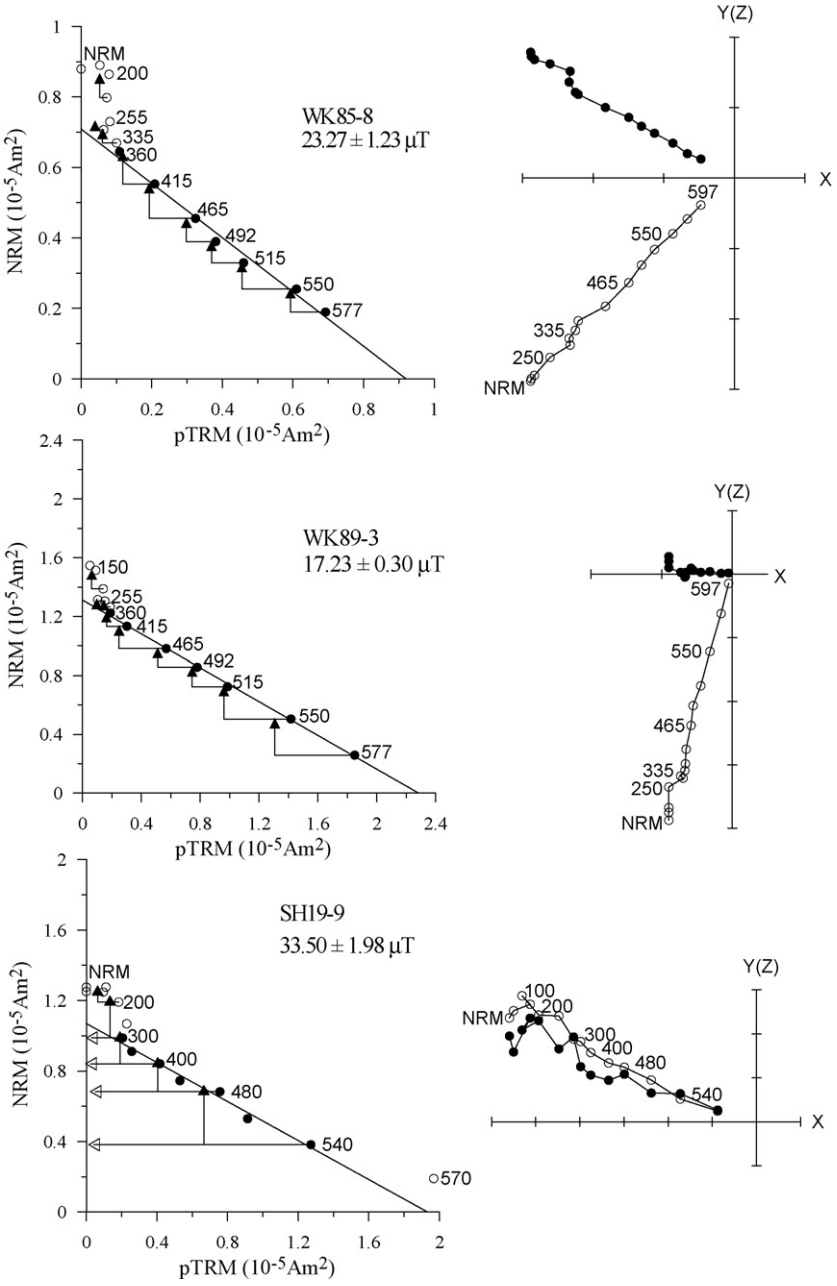


Fig. 7. (*Continued*).

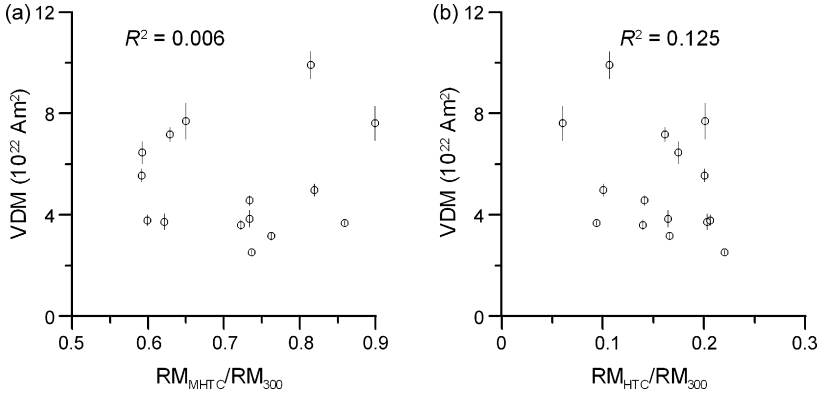


Fig. 8. Mean virtual dipole moments (VDMs) of lava flows or DGs versus their corresponding remanence components with different unblocking temperatures. RM_{-MHTC}, RM_{-HTC} and RM_{300} refer to intensities of the medium high-temperature component (300–585 °C), high-temperature component (585–630 °C) and initial ChRM, respectively.

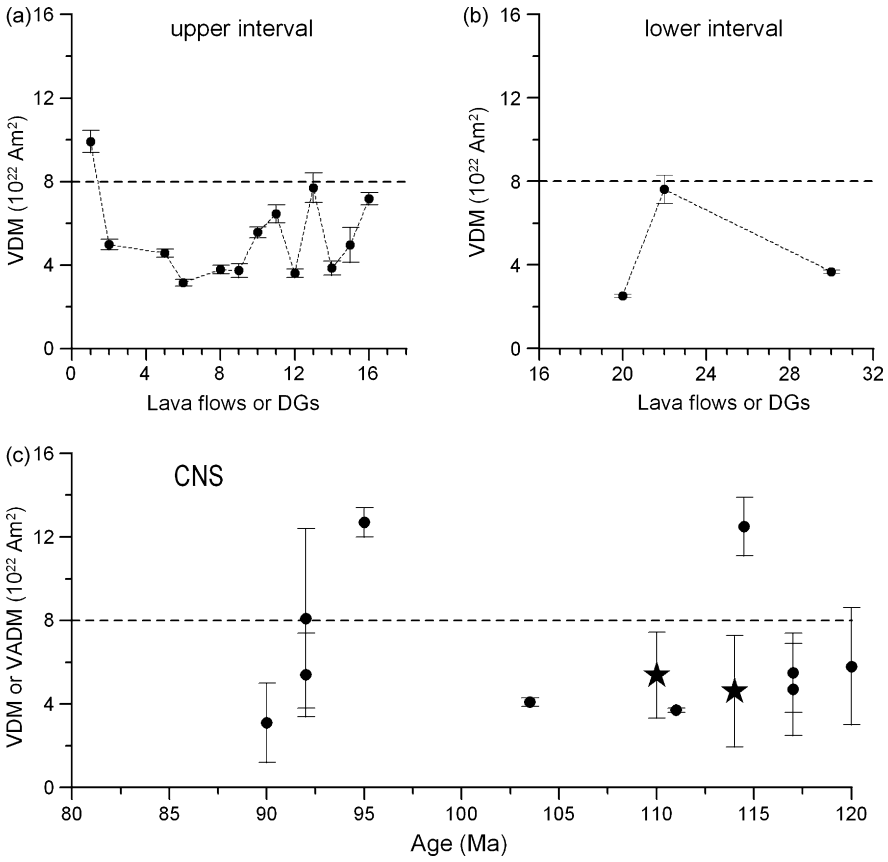


Fig. 9. Variation of virtual dipole moments (VDMs) determined from the upper interval, ~110 Ma (a) and lower interval, ~114 Ma (b) from the Suhongtu section, and comparisons of Thellier-type palaeointensity data within the Cretaceous normal superchron (CNS) (c) (see Table 3 for references). Dashed lines are the intensity of the present-day field. In (c), stars are two mean palaeointensities from this study.

As discussed above, >300 °C the ChRM was isolated and directions of MHTC (~300–585 °C) and HTC (585–630 °C) were almost the same (see Section 5). We further examined if the observed mean VDMs of lava flows are related to intensities of these two components having different unblocking temperatures to approximately evaluate the titanomagnetite and maghemite contributions. To this end, we analyzed the thermoremanences of 119 samples that yielded accepted palaeointensities. For each sample, the intensities of MHTC and HTC were calculated and named RM_{-MHTC} and RM_{-HTC}, respectively. The mean values of RM_{-MHTC} and RM_{-HTC} of lava flows were then normalized by its mean RM_{300}. On average, the RM_{-MHTC} component is dominant 72.0 ± 9.8%, while the RM_{-HTC} are 15.7 ± 4.7%. The remaining RM_{-HHTC} is 6.8 ± 3.5%. We compared the mean VDMs with these normalized RM_{-MHTC} or RM_{-HTC}, and found that the correlation is indeed very poor (Fig. 8). Specifically, the correlations between them (VDMs and RM component) are 0.006 and 0.125, respectively. In addition, there is no correlation between VDM values and measured hysteresis parameters (R^2 <0.15). Moreover, the excellent within-flow consistency of palaeointensity results is a strong defense against this and other concerns.

Therefore, we conclude that the present study on the consecutive Suhongtu section provides two palaeointensity time series during the CNS, namely around ~110 Ma and ~114 Ma (Fig. 9a and b). The most remarkable feature of the determined palaeointensities is the dramatic variation as a function of time (also

Table 3
Thellier palaeointensity data for the Cretaceous normal superchron (CNS)

No.	Location	Age (Ma)	Method	Lithology	VDM (VADM) 10^{22} Am^2	Std	Reference
1	NE China	90	K–Ar	WR	3.1	1.9	Tanaka and Kono (2002)
2	Troodos, Cyprus	~92		WR	5.4	2	Granot et al. (2007)
3	Troodos, Cyprus	~92		SBG	8.1	4.3	Tauxe and Staudigel (2004)
4	Arctic	95	^{40}Ar/^{39}Ar	SC	12.7	0.7	Tarduno et al. (2002)
5	NE China	102–105	^{40}Ar/^{39}Ar	WR	4.1	0.2	Zhu et al. (2004b)
6	Inner Mongolia	110	^{40}Ar/^{39}Ar	WR	5.4	2.1	This study
7	Nauru	111		SBG	3.7	0.1	Juarez et al. (1998)
8	Inner Mongolia	114	^{40}Ar/^{39}Ar	WR	4.6	2.7	This study
9	Rajmalhal Trap	113–116	^{40}Ar/^{39}Ar	SC	12.5	1.4	Tarduno et al. (2001)
10	Rajmalhal Trap	~117		WR	5.5	1.9	Sherwood et al. (1993)
11	Israel	~117		WR	4.7	2.2	Sherwood et al. (1993)
12	Ontong Java	~120		SBG	5.8	2.8	Riisager et al. (2003)

WR, SBG and SC refer to whole rock, submarine basaltic glass, and single crystal samples, respectively. VDM (VADM) and std, the calculated virtual dipole moment (virtual axial dipole moment) and its standard deviation.

see Table 2). The observed VDMs from the lower interval (three independent palaeomagnetic units) varied from 2.53×10^{22} Am^2 to 7.62×10^{22} Am^2 with a mean VDM of $(4.61 \pm 2.67) \times 10^{22}$ Am^2, while from the upper interval (12 independent palaeomagnetic units) varied from 3.16×10^{22} Am^2 to 9.92×10^{22} Am^2 with a mean VDM of $(5.38 \pm 2.06) \times 10^{22}$ Am^2. Granot et al. (2007) also reported variable palaeointensity during the CNS (around ~92 Ma), thus, we propose that fast fluctuating field intensity is a feature of the CNS. These fluctuations are similar to geomagnetic intensity variations seen over the past 780 kyr as recorded by sea-floor magnetic anomalies (e.g., Gee et al., 2000), marine sediments (e.g., Guyodo and Valet, 1999), and terrestrial lava flows (e.g., Teanby et al., 2002). Recently, Johnson et al. (2008) documented that the Brunhes and Matuyama chrons exhibit different time-averaged field geometries and palaeosecular variation behavior. Notable smaller S values observed at the Suhongtu section partly suggest that departures from a geocentric axial dipole field during the CNS be smaller than during the Brunhes chron, which results in a fast variation of dipole strength with stable polarity.

We further compare the palaeointensity results of this study with the published Thellier palaeointensity data (with pTRM checks) for the CNS in Table 3 and Fig. 9c. Palaeointensity from the Troodos Ophiolite (~92 Ma) is comparable to the present-day field (Tauxe and Staudigel, 2004). Determinations from the High Canadian Arctic lavas (~95 Ma) and the Rajmalhal trap (113–116 Ma) using single crystals, which it is suggested have suffered the least magnetic mineral alteration in nature, yielded higher VDMs ($\sim 12 \times 10^{22}$ Am^2) (Tarduno et al., 2001, 2002). Data of whole rock samples are in general lower compared with the present-day field. On the other hand, lower palaeointensity observed just prior to and after the CNS might support the prediction of positive relationship between the length of polarity interval and the dipole intensity (e.g., Zhu et al., 2001, 2003; Pan et al., 2004; Tarduno et al., 2006).

8. Conclusions

The Early Cretaceous basaltic-andesitic lava sequence at the Suhongtu section recorded two short time-series variations of the geomagnetic field around 114 Ma and 110 Ma, inside the CNS. The palaeomagnetic pole inferred from 31 independent palaeomagnetic directions, all having normal polarities, is consistent with the Eurasian pole determined from the APWP curve of the same age, suggesting that primary remanent magnetization were preserved and no significant tectonic movement has occurred since lava emplacement. High quality palaeointensity data demonstrate that the field intensity dramatically fluctuated (varying from 2.53×10^{22} Am^2 to 9.92×10^{22} Am^2). The mean VDMs for the upper and lower intervals are $(5.38 \pm 2.06) \times 10^{22}$ Am^2 and $(4.61 \pm 2.67) \times 10^{22}$ Am^2, respectively. Taken together with available global data within the CNS, we infer that the field fluctuated during the superchron.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.pepi.2008.07.025.

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